

IPsec overlapping subnets in Palo Alto

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IPsec overlapping subnets in Palo Alto

Date: 06-18-2026

Overlapping subnets: when the remote subnets are the same, we use the term called as overlapping subnets. To overcome this, we use virtual subnets, which don't exist in both remote and local interfaces. Virtual subnets will sit between local and remote subnets and do the NAT translation in between.

For Example:

Local Subnet: 192.168.4.0/24

Remote Subnet: 192.168.4.0/24

So when local subnets exist the same as remote subnets, we cannot tell the router or routing table to route to the Tunnel and LAN interface for the same subnet, so we need another duplicate subnet that doesn't exist to use it.

Steps to create an IPsec tunnel:

Site -1:

1. Login into the Palo Alto GUI page

Address Objects

2. Navigate to Address objects and create the following address objects

Site -1:

Local Subnet: 192.168.4.0/24

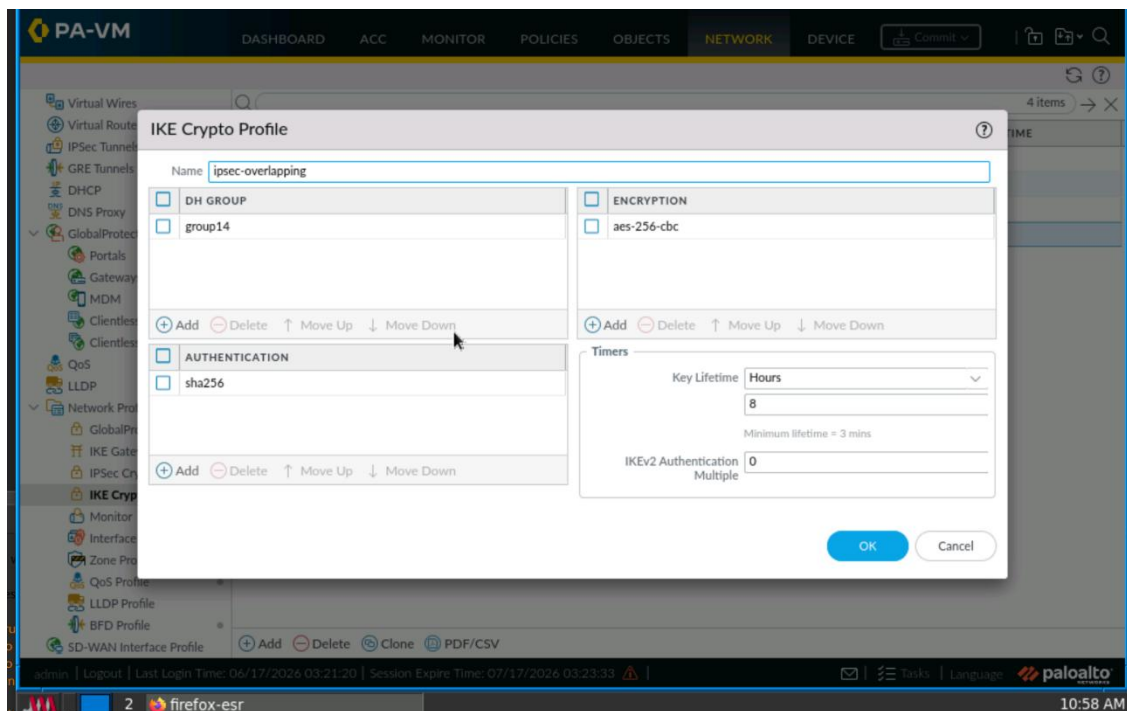
Local Translated Subnet: 10.1.10.0/24

Remote Translated Subnet: 10.1.20.0/24

IPsec tunnel:

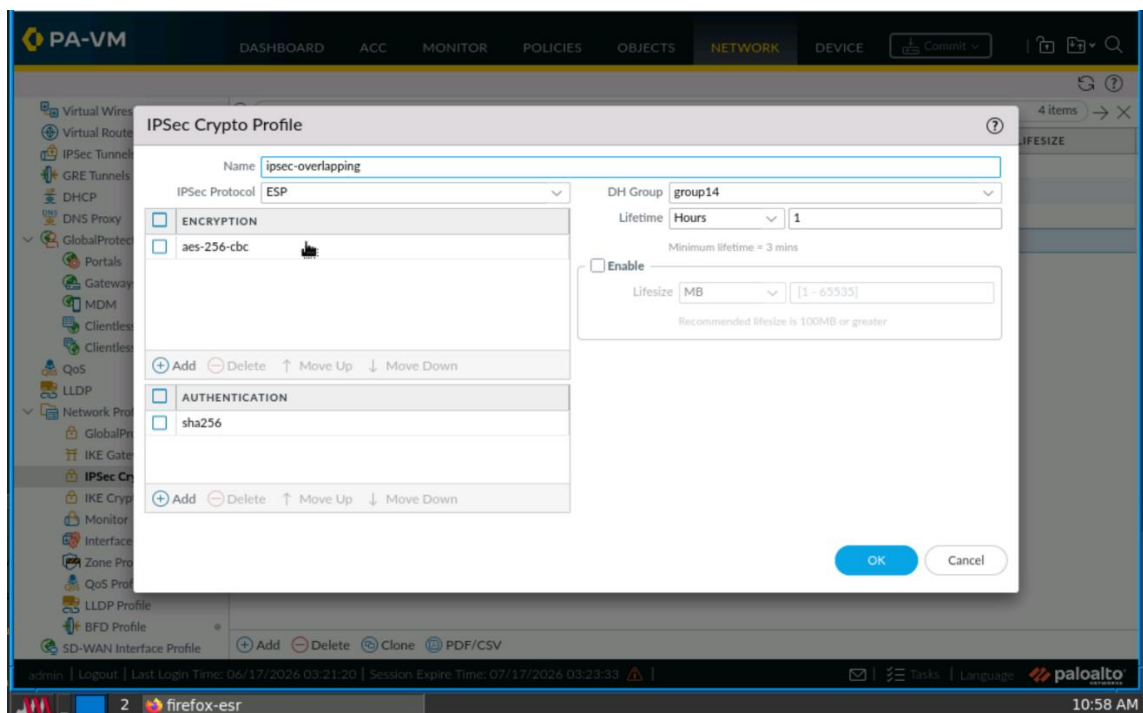
3. Navigate to Network > IKE Crypto
4. Create a IKE profile (Phase -1) with secure authentication, DH Group and encryption

Note: Both sites should have the same properties to agree on the proposal and to establish a tunnel



5. Create IPsec Crypto Profile(Phase-1) with Authentication, DH Group and Encryption

Note: Both sites should have the same properties to agree on the proposal and to establish a tunnel



6. Create an IKE Gateway with the IKE Profile and Crypto profiles, outbound interface, and pre-shared key.

Choose the preferred mode that allows flexibility for Palo between Ikev1 and Ikev2 based on peer negotiation

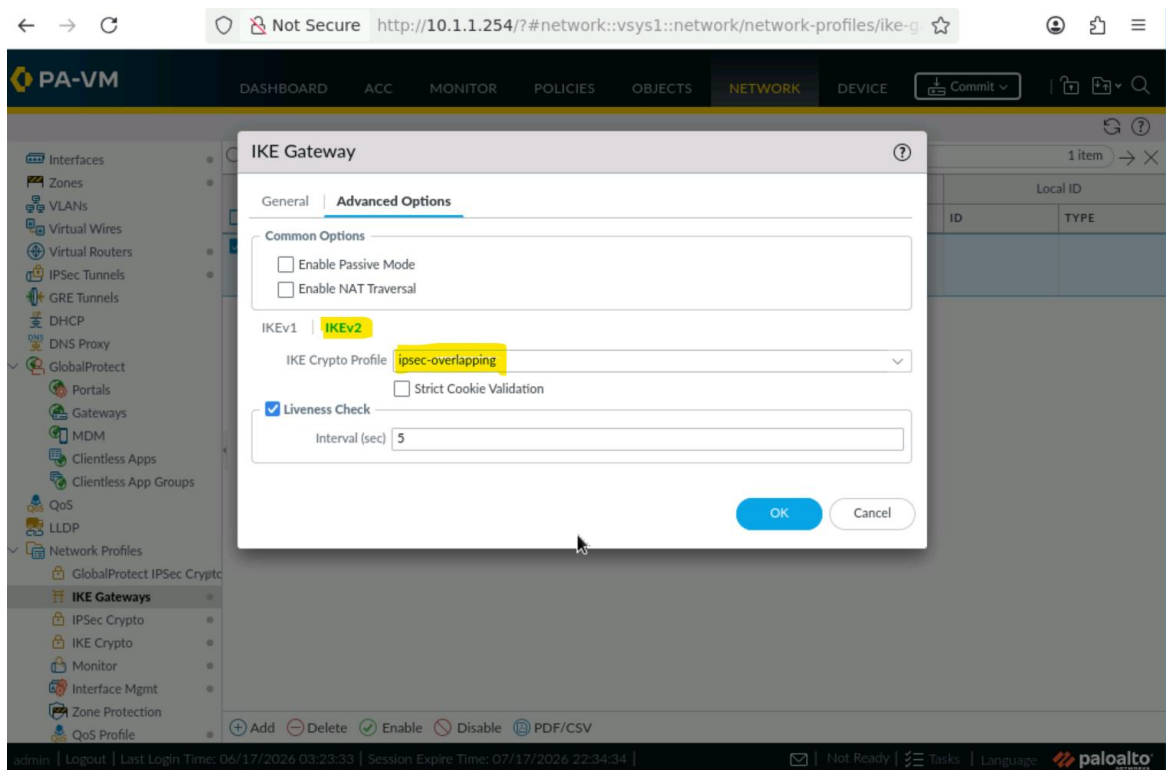
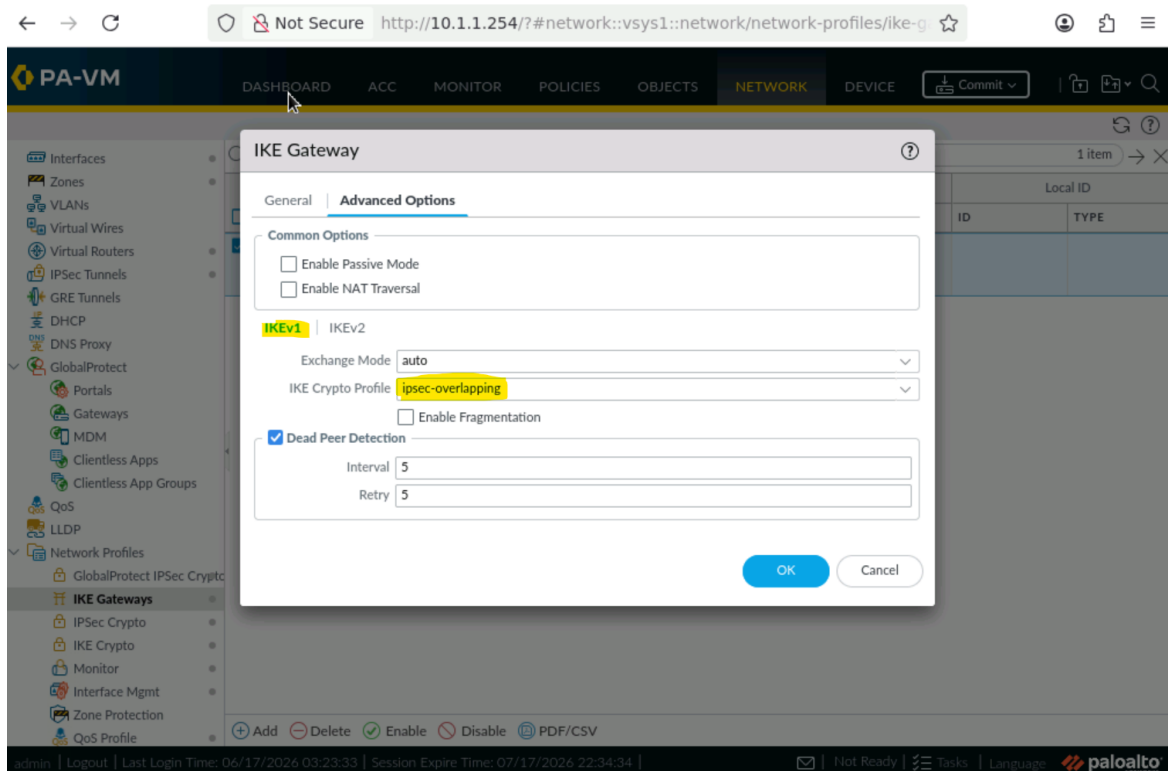
Choose the Interface which is Outbound(WAN interface and IP should be reachable to the peer)

Provide the Peer IP in the Peer address

The screenshot shows the Palo Alto VM configuration interface with the 'IKE Gateway' dialog box open. The dialog has two tabs: 'General' and 'Advanced Options'. The 'General' tab is active, showing the following configuration:

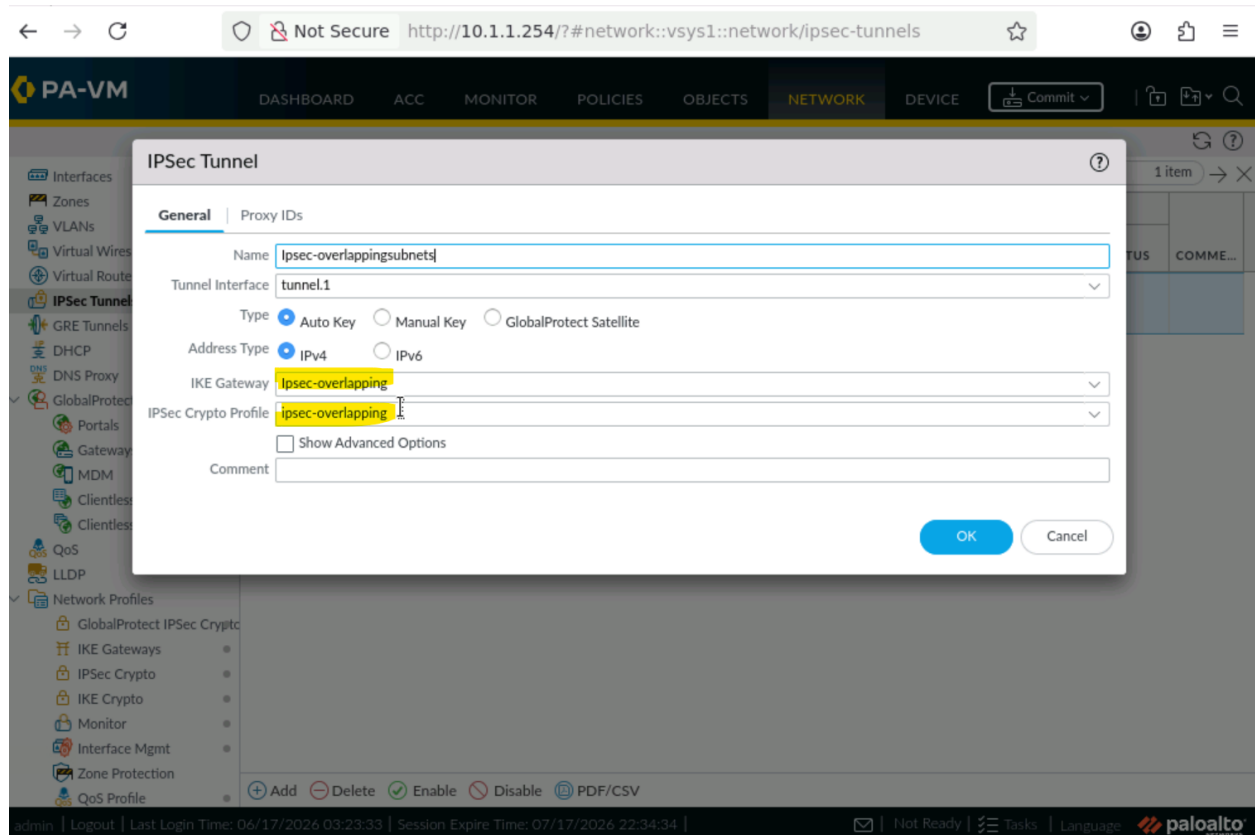
- Name:
- Version:
- Address Type: ☒ IPv4 ☐ IPv6
- Interface:
- Local IP Address:
- Peer IP Address Type: ☒ IP ☐ FQDN ☐ Dynamic
- Peer Address:
- Authentication: ☒ Pre-Shared Key ☐ Certificate
- Pre-shared Key:
- Confirm Pre-shared Key:
- Local Identification:
- Peer Identification:
- Comment:

The dialog has 'OK' and 'Cancel' buttons at the bottom right. The background shows the Palo Alto VM configuration interface with the 'NETWORK' tab selected. The left sidebar shows the configuration tree with 'IKE Gateways' highlighted. The bottom status bar shows the user 'admin', login time '06/17/2024 03:21:20', session expire time '07/17/2024 03:23:33', and the Palo Alto logo.



Note: Enable NAT Traversal only when the device is behind the router or Nat device

7. Navigate to IPsec tunnels and create an IPsec tunnel with the IKE Gateway

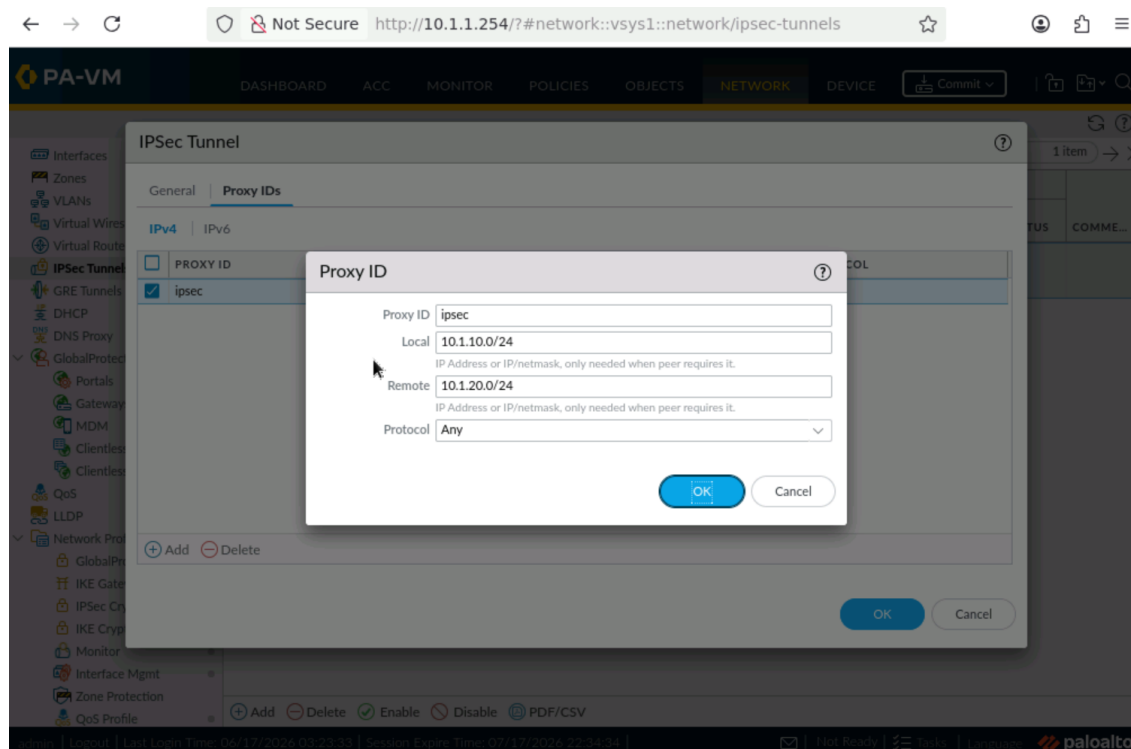


Click on proxy IDs and add proxy IDs (here is where we inform local and remote subnets to the IPsec tunnel)

Local Subnet should be the Local Translated: 10.1.10.0/24

Remote Subnet should be the remote translated: 10.1.20.0/24

Choose protocol: Any



Create Zone and Tunnel Interface

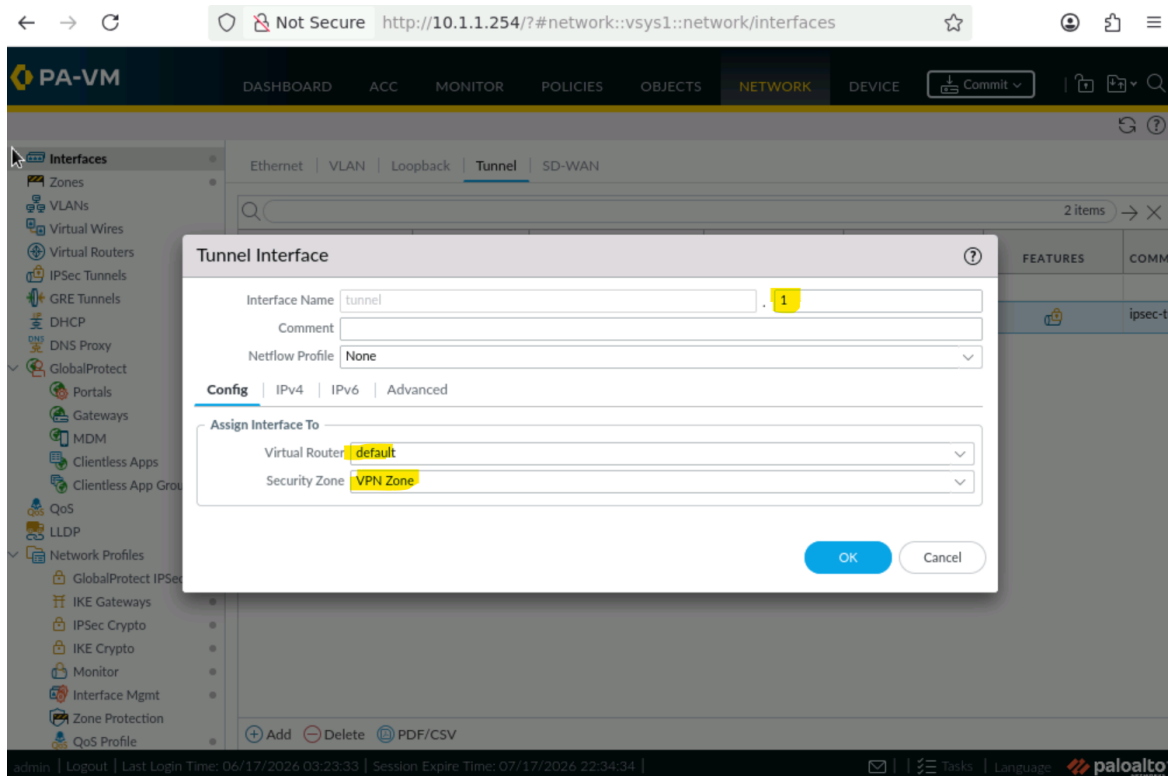
8. Navigate to Network> Interfaces and click on the tunnel tab

Click on add

Choose the Virtual router(mostly the default one)

Click on Security Zone > New Zone and create a Layer 3 Zone name as a VPN

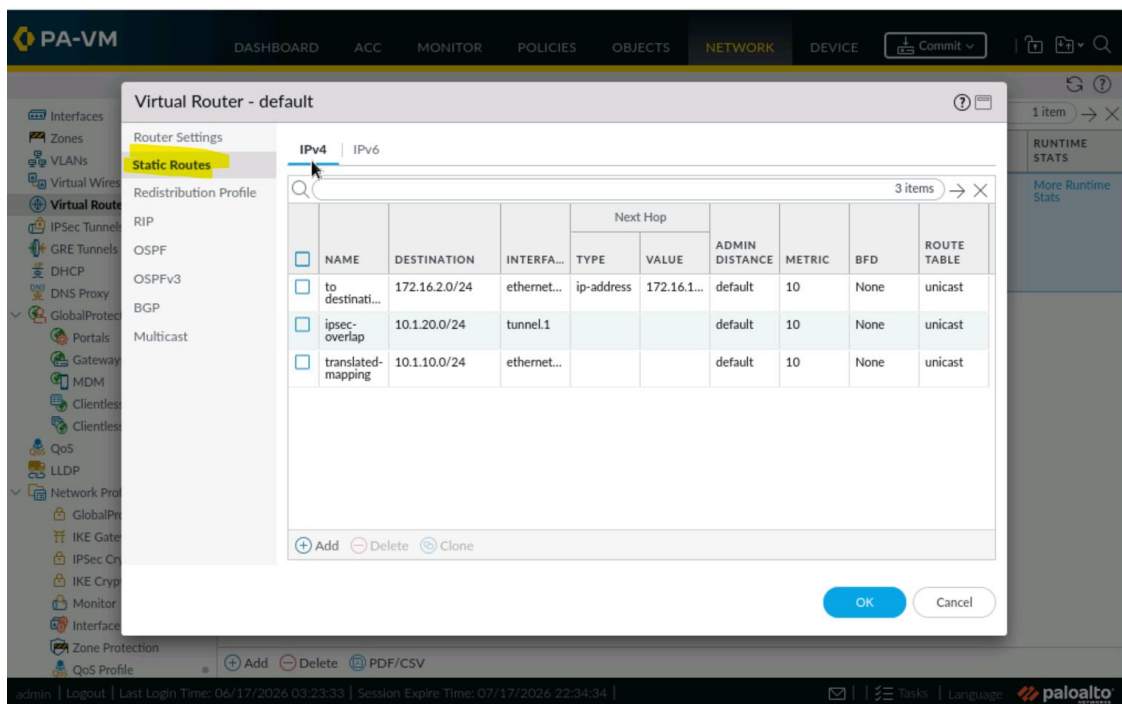
Choose the Tunnel interface as a unique number, such as 1, 2, or 3, etc.



Create Static Routes

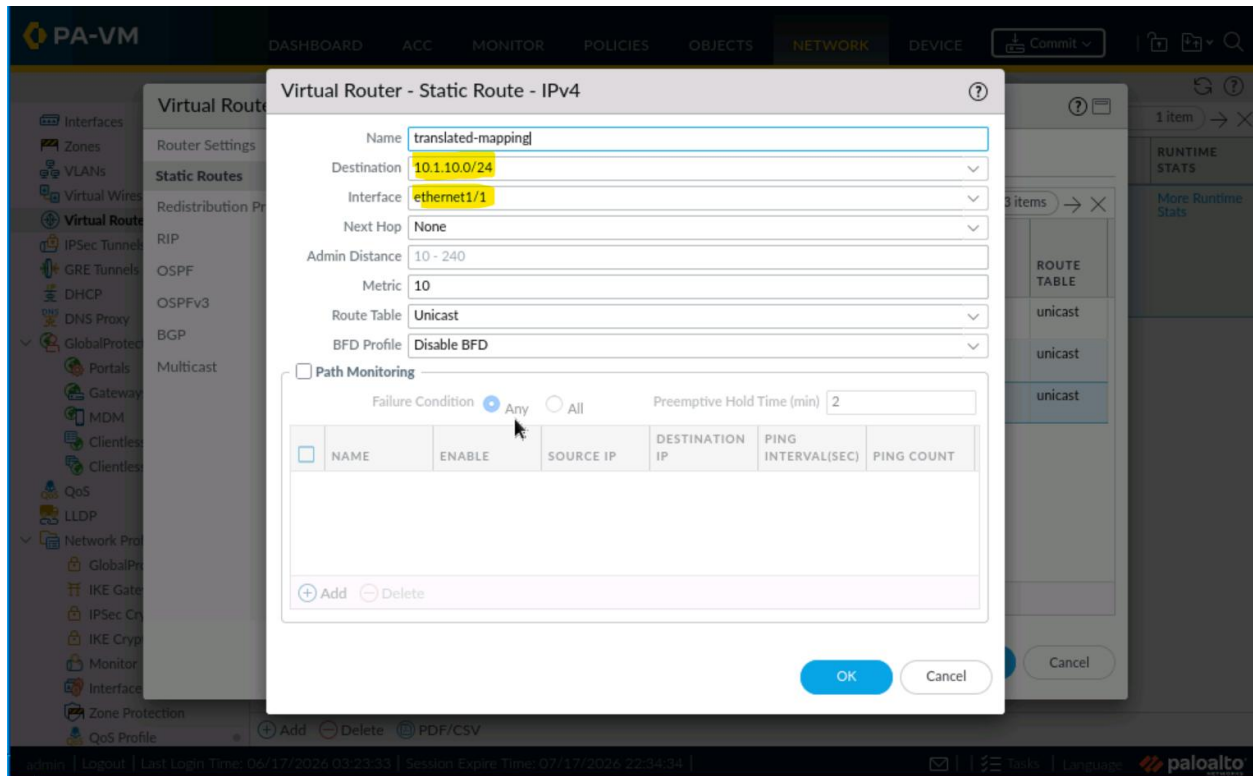
9. Navigate to Network > Virtual routers

Edit the Default



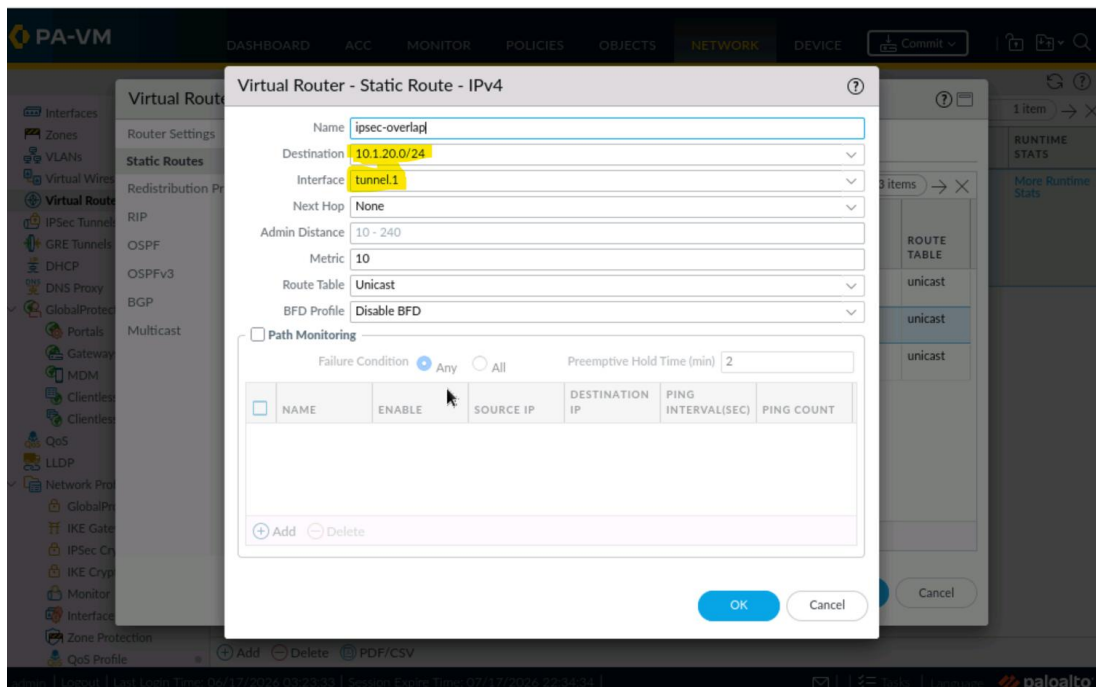
Click on Add to add the static routes

We need to add the local translated address to point to LAN interface where original local subnets exists this helps routing table to know where this subnets exists if packets received with the destination as 10.1.10.0/24 from peer.



We also need to add the remote translated subnet as a static route this helps the routine engine (RE) to send traffic via specific interface when the packets arrive with that subnet

Click on add and add another static pointing to tunnel interface we created (point .8)



Create SNAT and DNAT policies

10. Navigate to Policies > NAT

SNAT

Click on add

Name the policy as SNAT_to_Site_B

In the original Packet section

Select the source zone as the LAN zone where the original subnet exists

Select the destination Zone as the VPN zone where the remote subnets exist

Source Address: Local subnet(original) – 192.168.4.0/24

Destination Address: Remote Translated Address – 10.1.20.0/24

In the translated packet section

Source Address Translation Section

Translation type: Static IP

Translated Address: Local Translated (10.1.10.0/24)

Enable Bi-direction

Click on Ok

PA-VM

DASHBOARD ACC MONITOR **POLICIES** OBJECTS NETWORK DEVICE Commit

Security NAT QoS Policy Based Decryption Tunnel Inspection Application Control Authentication DoS Protection SD-WAN

Policy Optimizer Rule Usage Unused Unused Unused

NAT Policy Rule

General **Original Packet** Translated Packet

☐ Any	Destination Zone	☐ Any	☐ Any
☒ SOURCE ZONE ^	VPN Zone	☒ SOURCE ADDRESS ^	☒ DESTINATION ADDRESS ^
☒ LAN	Destination Interface	☒ LAN-Subnet	☒ translated ip
	any		
	Service		
	any		
+ Add - Delete		+ Add - Delete	+ Add - Delete

Object : Addresses + Move PDF/CSV Highlight Unused Rules View Rulebase as Groups

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DASHBOARD ACC MONITOR **POLICIES** OBJECTS NETWORK DEVICE Commit

Security NAT QoS Policy Based Decryption Tunnel Inspection Application Control Authentication DoS Protection SD-WAN

Policy Optimizer Rule Usage Unused Unused Unused

NAT Policy Rule

General **Original Packet** **Translated Packet**

Source Address Translation	Destination Address Translation
Translation Type Static IP	Translation Type None
Translated Address translated	
☒ Bi-directional	

Object : Addresses + Move PDF/CSV Highlight Unused Rules View Rulebase as Groups

admin | Logout | Last Login Time: 06/17/2026 03:23:33 | Session Expire Time: 07/17/2026 22:34:34 | Tasks Language paloalto

DNAT

Click on add

In the original packet section

Source Zone: VPN

Destination Zone: LAN

Source Address: Remote Translated Address(10.1.20.0/24)

Destination Address: Local Translated Address(10.1.10.0/24)

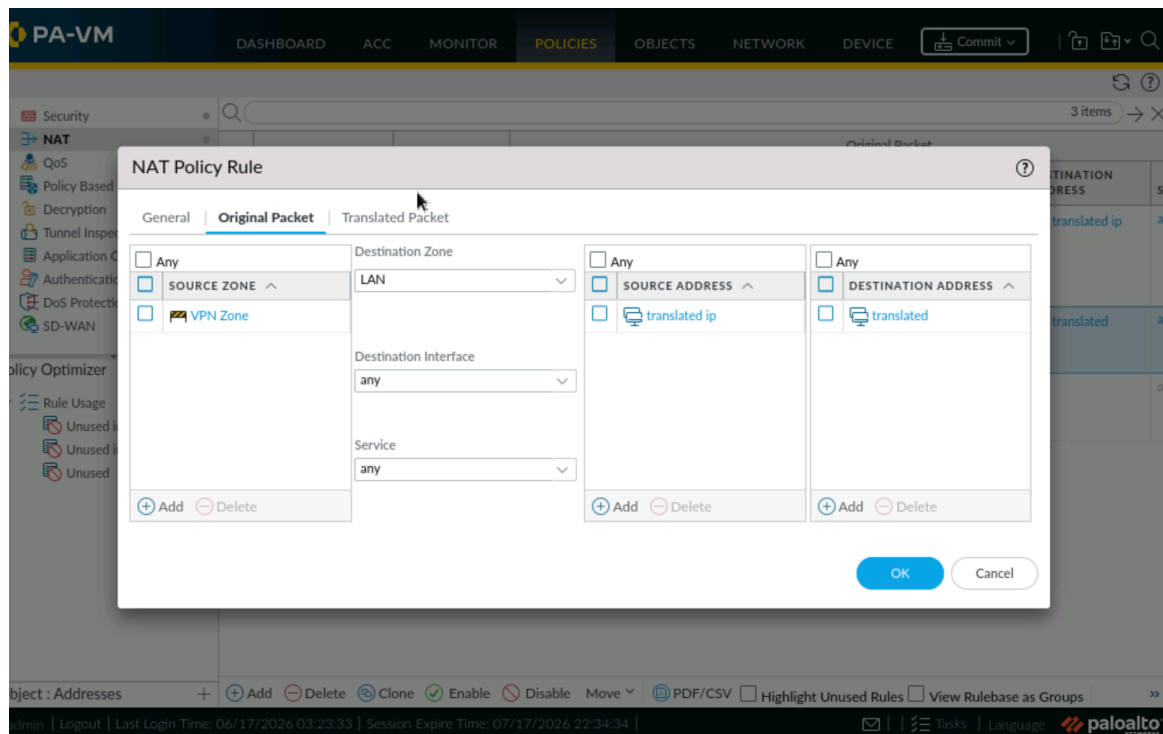
In the translated packet section

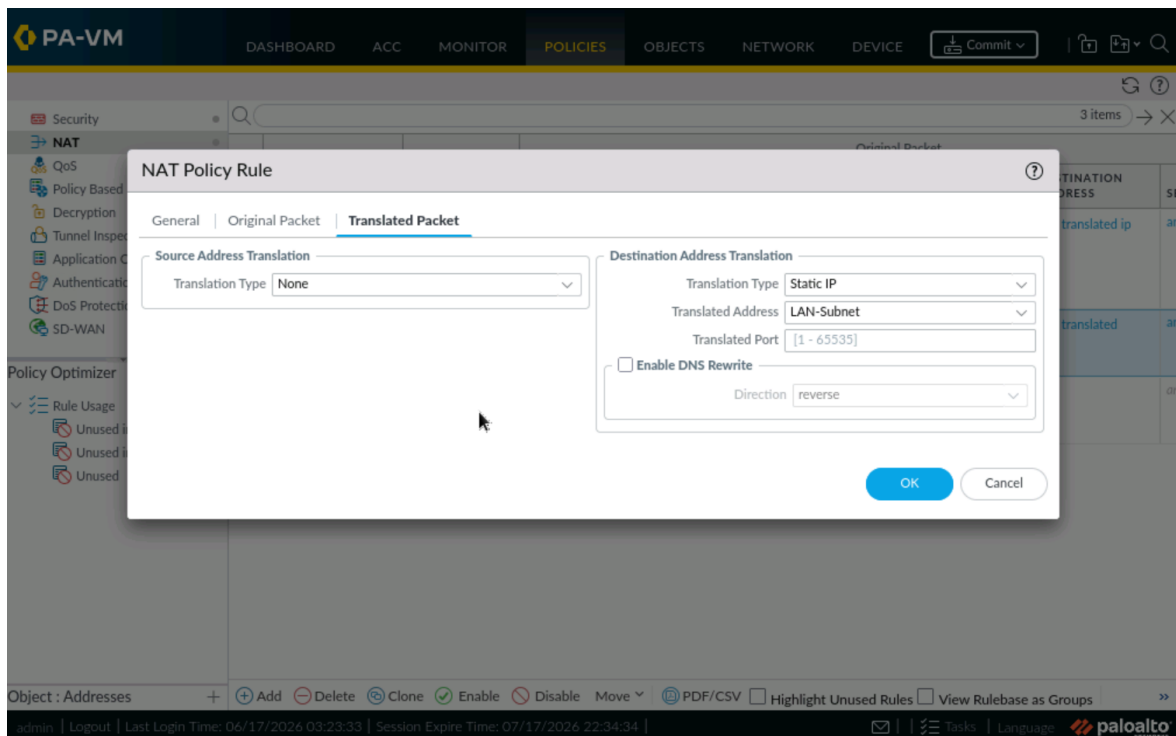
Destination Address Translation section

Translation Type: Static IP

Translated Address: LAN-Subnet

Click ok to save to the candidate configuration





Create Security Policy

11. Navigate to Policies > Security

We have to create two policies, one for the site_a > site_b and site_b > site_a

site_a > site_b

Click on add

Name the policy as Site_A to Site_B

Source Tab

Source Zone: LAN

Source Address: LAN Subnet(192.168.4.0/24)

Destination Tab

Source Zone: VPN

Destination address: Remote Translated IP (10.1.20.0/24)

Application tab: Any or restrict traffic based on the requirement

Actions tab: Allow

PA-VM DASHBOARD ACC MONITOR **POLICIES** OBJECTS NETWORK DEVICE Commit

Security 5 items

NAT

Security Policy Rule

General **Source** Destination Application Service/URL Category Actions Usage

☐ Any	☐ Any	any	any
☐ SOURCE ZONE ^	☐ SOURCE ADDRESS ^	☐ SOURCE USER ^	☐ SOURCE DEVICE ^
☐ LAN	☐ LAN-Subnet		
+ Add - Delete	+ Add - Delete	+ Add - Delete	+ Add - Delete

☐ Negate

OK

Object : Addresses + Add - Delete Clone Override Revert Enable Disable Move PDF/CSV Highlight Unused Rules

admin | Logout | Last Login Time: 06/17/2024 03:23:33 | Session Expire Time: 07/17/2024 22:34:34 | Tasks Language **paloalto**

PA-VM DASHBOARD ACC MONITOR **POLICIES** OBJECTS NETWORK DEVICE Commit

Security 5 items

NAT

Security Policy Rule

General Source **Destination** Application Service/URL Category Actions Usage

select	☐ Any	any
☐ DESTINATION ZONE ^	☐ DESTINATION ADDRESS ^	☐ DESTINATION DEVICE ^
☐ VPN Zone	☐ translated ip	
+ Add - Delete	+ Add - Delete	+ Add - Delete

☐ Negate

OK

Object : Addresses + Add - Delete Clone Override Revert Enable Disable Move PDF/CSV Highlight Unused Rules

admin | Logout | Last Login Time: 06/17/2024 03:23:33 | Session Expire Time: 07/17/2024 22:34:34 | Tasks Language **paloalto**

site_b > site_a

click on Add

Name the policy as site_a > site_b

Source Tab

Source Zone: VPN

Source Address: Remote translated address(10.1.20.0/24)

Destination Tab

Source Zone: LAN

Destination Address: Local Translated Address(10.1.10.0/24)

Application tab: Any or restrict traffic based on the requirement

Actions tab: Allow

The screenshot shows the Palo Alto VM configuration interface for a Security Policy Rule. The interface is divided into several tabs: General, Source, Destination, Application, Service/URL Category, Actions, and Usage. The Source tab is currently selected, showing four configuration sections: SOURCE ZONE, SOURCE ADDRESS, SOURCE USER, and SOURCE DEVICE. Each section has a list of items and an 'Add' button. The SOURCE ZONE section shows 'VPN Zone' selected. The SOURCE ADDRESS section shows 'translated ip' selected. The SOURCE USER and SOURCE DEVICE sections are empty. Below these sections is a 'Negate' checkbox. At the bottom right, there is an 'OK' button. The bottom of the interface shows a status bar with the object name 'Addresses', a list of actions (Add, Delete, Clone, Override, Revert, Enable, Disable, Move), and a 'Highlight Unused Rules' checkbox. The Palo Alto logo is visible in the bottom right corner.

PA-VM

DASHBOARD ACC MONITOR **POLICIES** OBJECTS NETWORK DEVICE

Commit

Security

NAT

Security Policy Rule

General **Source** Destination Application Service/URL Category Actions Usage

☐ Any

☐ SOURCE ZONE ^

☐ VPN Zone

☐ Add ☐ Delete

☐ Any

☐ SOURCE ADDRESS ^

☐ translated ip

☐ Add ☐ Delete

☐ Any

☐ SOURCE USER ^

☐ Add ☐ Delete

☐ Any

☐ SOURCE DEVICE ^

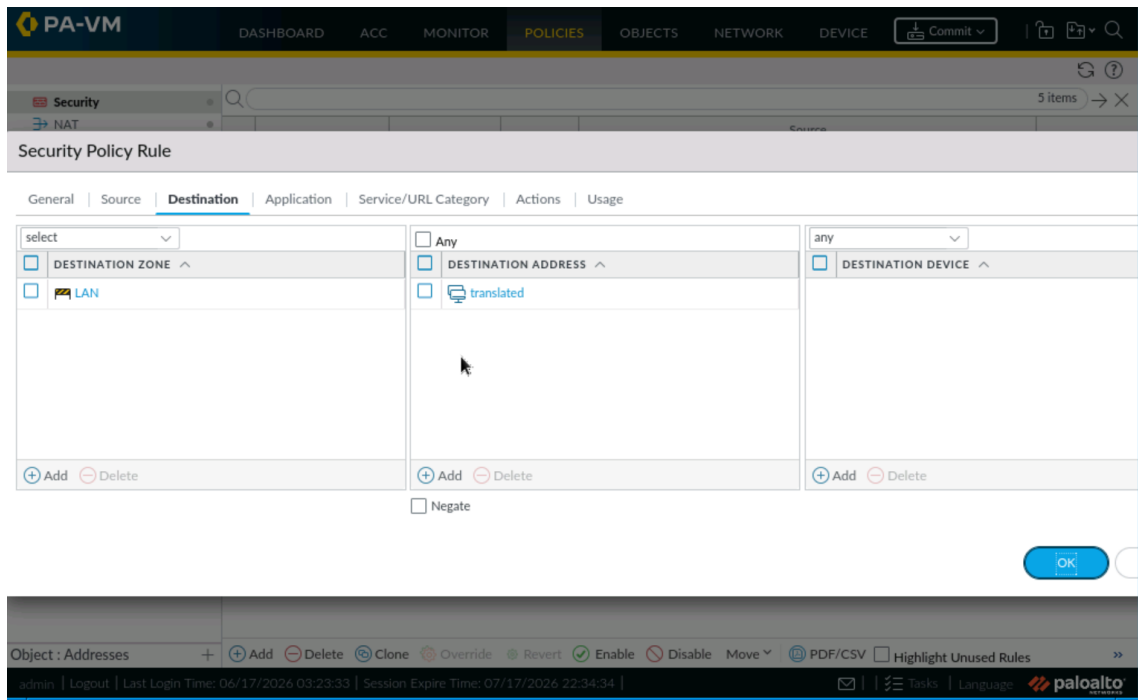
☐ Add ☐ Delete

☐ Negate

OK

Object : Addresses + ☐ Add ☐ Delete ☐ Clone ☐ Override ☐ Revert ☒ Enable ☐ Disable ☐ Move PDF/CSV ☐ Highlight Unused Rules

admin | Logout | Last Login Time: 06/17/2026 03:23:33 | Session Expire Time: 07/17/2026 22:34:34 | Tasks | Language | paloalto



Repeat the Exact Opposite on the Remote Peer

Local Subnet: 192.168.4.0/24

Local Translated Address: 10.1.20.0/24

Remote Translated Address: 10.1.10.0/24

How the Translation Happens

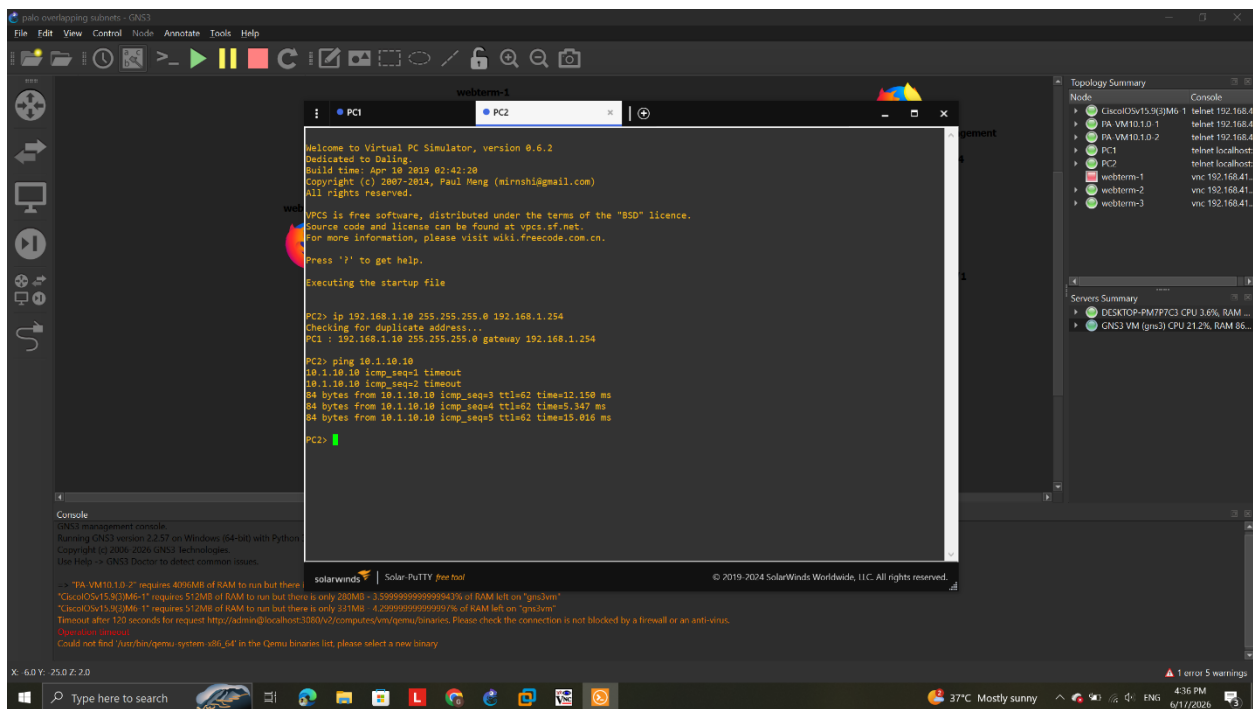
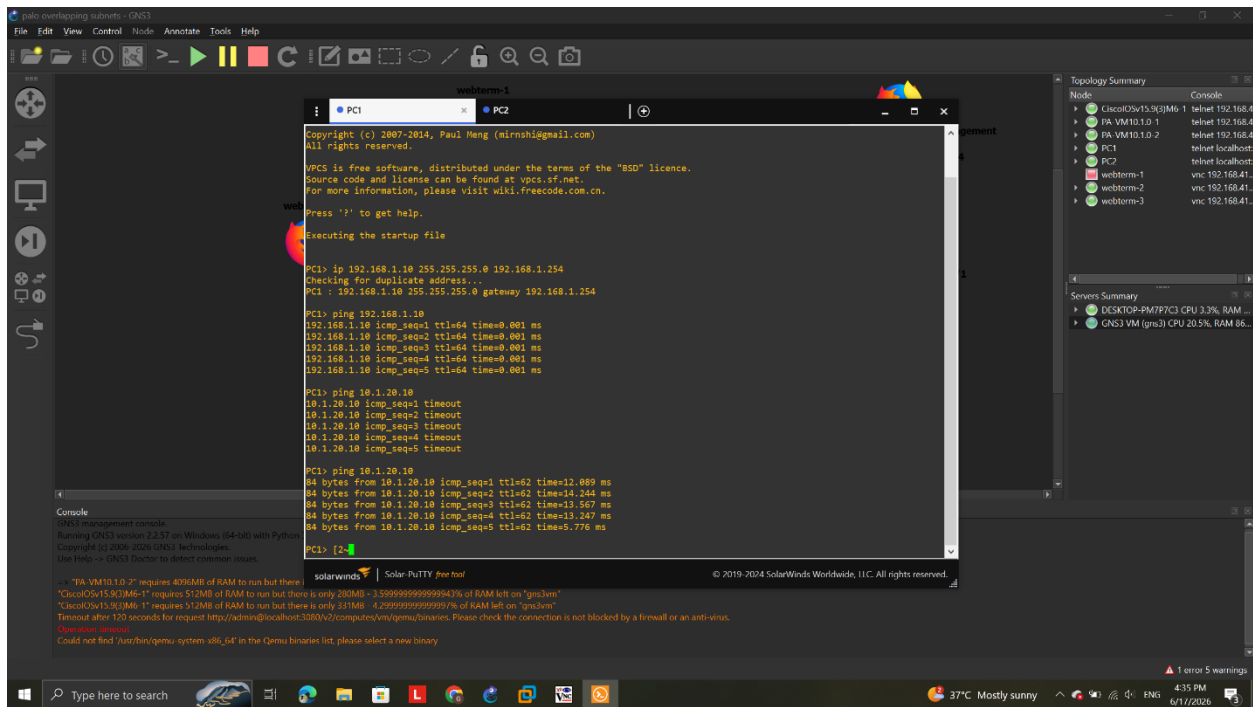
Let's say if the PC1 in Site A has the IP address 192.168.1.10 and it will SNAT 192.168.4.10 < 10.1.10.10

Site B firewall sees the packet source field as 10.1.10.10 and destination field as 10.1.20.10 it will DNAT to 10.1.20.10 < 192.168.4.10

Test:

Take a PC1 in Site A and ping any IP in the remote subnets, you will get a response.

The first two packets will be lost because if the tunnel is down, one loss makes the tunnel up, and the second loss builds the ARP.



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